

You don't always need LLM!

The best AI Engineers know when to use what

Here's a simple roadmap to guide you 📌

If you're serious about NLP (basic → LLMs),
these are the only books you need to backup your theory.

NLP (basic to LLM) roadmap:

- ❶ **Python Natural Language Processing Cookbook** - Zhenya Antić, PhD & Saurabh Chakravarty, Ph.D.
 - ↳ Tokenization, n-grams, stop-words, stemming, TF-IDF, BM25, NER.
 - ↳ Hands-on with NLTK, spaCy, PyTorch, Hugging Face.
 - ↳ word2vec, GloVe, FastText (pre-transformer must-knows).

- ❷ **Natural Language Processing with Transformers** - Lewis Tunstall, Leandro von Werra, Thomas Wolf
 - ↳ From BERT & DistilBERT to sentence-transformers.
 - ↳ Hugging Face workflows for classification, QA.
 - ↳ Super clear on transformer internals.

- ❸ **Hands-On Large Language Models** - Jay Alammar & Maarten Grootendorst
 - ↳ Build a transformer from scratch.
 - ↳ Prompting, embeddings, RAG, eval.
 - ↳ Function calling, modern agent pipelines.

Possibly you could also add - Build a Large Language Model (From Scratch) by Sebastian Raschka, PhD

That's really it!

(there would be some overlap of topics across the books)

These books are really enough for your theoretical backing - further try to incorporate these principles in your hands-on projects.